

When Sample Selection Bias Precipitates Model Collapse

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Introduction

Recursive training on generated data can narrow distributional support when later generations are selected by biased local references. The key question is when accumulation preserves diversity, and when selection bias overrides it.

Theory

Replace paradigm [R1]
Replacing data by the latest generated round drives covariance loss and tail erosion.
Accumulate paradigm [R2]
Pooling generations can stabilize unbiased recursion, but biased selection changes the limit.
Theorem 1
Under top-alpha selection toward a local target, the mean approaches that target while covariance contracts.
Theorem 2
Within the local basin, the covariance trace follows a power-law decay, giving an explicit collapse rate.
Theorem 3
After fitting error is small, target-domain risk is governed by distributional mismatch.

References

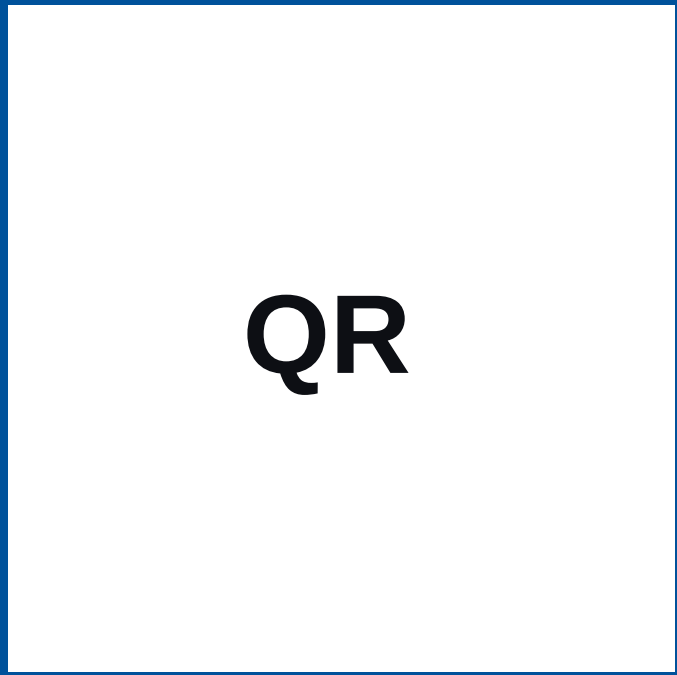
[R1] Replace-paradigm model-collapse analysis. [R2] Accumulate-paradigm recursive training analysis.

For more methodology and theoretical details, please refer to the full paper.

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Scan
OpenReview paper and supplement.
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Selection bias
accelerates collapse.
Accumulation slows
but does not prevent it.
Scaling sample count
does not restore diversity.

Multivariate Gaussian Modeling

Gaussian
Selection contracts covariance under both replacement and accumulation.
Anisotropic Gaussian
Directional variance changes the path, not the final loss of dispersion.
Gaussian mixture
Biased filtering reduces mode coverage and tail mass across generations.
Laplace
Heavy-tailed initialization still loses dispersion under biased filtering.
Trace ratio
$\text{Tr}(\Sigma_t)/\text{Tr}(\Sigma_0)$ summarizes diversity contraction. Replace collapses fastest; Accumulate decays more slowly.

Empirical Summary

Across Gaussian and non-Gaussian families, biased selection is the dominant driver of diversity loss and tail-coverage erosion.

For additional experimental results, please refer to the full paper.

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